

Big Data

Degree in Data Science and Engineering

School of Computer Engineering

Academic year 2026–2027

Individual Assignment 2

Optimized Approaches and Sparse Matrices

How much do cache-aware optimisations speed up dense matrix multiplication,
and when does a sparse representation pay off?

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Repository:

https://github.com/Claudia1771/bigdata-matrix-multiplication

(folder `assignment2`)

October 28, 2026

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Abstract

This study extends the basic triple-loop matrix multiplication of Assignment 1 in two directions, using our own C implementations, single-threaded runs pinned to one performance core, and five repetitions summarised by the median and the interquartile range. In Part A, three optimisations of the validated baseline that keep exactly the same floating-point operations (loop reordering, transposition and cache blocking with an experimentally chosen block of 64×64) were compared for $n = 256$ to 4096. All of them were faster at every size; cache blocking was the fastest for large matrices, reaching 1.66 s for $n = 2048$ against 86.6 s for the baseline (52 times faster) and completing $n = 4096$ in 15.3 s, where the baseline exceeded the 300 s limit. Its cost per multiply-add stayed almost constant (0.17–0.22 ns), so its time followed the cubic model, and the power-of-two penalty of the baseline (4.3 times slower at $n = 1024$ than at $n = 1000$) practically disappeared. In Part B, a CSR implementation of Gustavson's algorithm was compared with the dense blocked product of the same matrices for densities from 0.01% to 50% and two structures. With uniformly random nonzeros the sparse product was faster below a break-even density of about 44% (43% including the conversion from dense), but its advantage dropped quickly because the product fills in (an input density of 5% already gives a 99% dense result); with banded nonzeros it was still faster at 50%. For a density of 0.1% the sparse product handled $n = 32768$ in 2.1 s and 426 MiB, while the dense product needed 110 s and 1.5 GiB at $n = 8192$ and could not run beyond it within the 4 GiB budget. The results are limited to the tested implementations, inputs and machine.

1. Introduction and Objective

Assignment 1 showed that the basic triple-loop matrix multiplication follows the cubic model only while its data fits in the caches, and that its cost is dominated by the memory access pattern of the innermost loop, which reads a column of B [1]. This study builds on that result in two directions. Part A keeps the dense problem and changes only the order in which the same arithmetic is performed, to make better use of the memory hierarchy [5, 6]. Part B changes the representation of the data: when most entries are zero, a sparse format stores and multiplies only the nonzeros [3, 4], but the benefit depends on how many nonzeros there are, where they are, and how dense the product becomes.

1.1. Research questions

- **RQ-A.** How much do loop reordering, transposition and cache blocking reduce the time of the basic dense multiplication of Assignment 1, and how does the gain change with the matrix size?
- **RQ-B.** Under which tested densities and nonzero structures does a CSR sparse multiplication outperform a dense multiplication of the same matrices, once the conversion cost and the density of the result are considered?

The factors changed are the method, the matrix size n (side of an $n \times n$ matrix), the block size (tiled method), and, in Part B, the density (percentage of nonzero entries; sparsity $= 100-$ density) and the structure of the nonzeros. The outcomes are the execution time of one multiplication, the peak memory of the process, and, in Part B, the number of nonzeros of the result.

1.2. Hypotheses

Stated before running the experiments:

H1. All three optimisations are faster than the baseline, and their advantage grows with n, because the baseline degrades as its working set leaves the caches while sequential access does not.

H2. Cache blocking is the fastest dense method for the largest sizes, because it reuses blocks of all three matrices while they are in the cache.

H3. The optimisations remove the power-of-two penalty observed in Assignment 1 ($n = 1024$ much slower than $n = 1000$).

H4. The CSR product is faster than the dense product only below some density; for

uniformly random nonzeros the result fills in quickly, so this break-even density is lower than for banded matrices with the same density.

H5. For very sparse matrices, the CSR product makes sizes feasible that are impossible in dense form within the memory budget.

2. Background

2.1. Dense optimisations

The baseline is the i-j-k loop of Assignment 1 (Algorithm 1), whose innermost loop reads B by columns with a stride of n elements. Three classic transformations keep exactly the same n^3 multiply-adds:

- **Loop reordering (i-k-j).** Exchanging the two inner loops makes the innermost loop update a row of C with a scalar multiple of a row of B, so both are traversed sequentially [7, 8].
- **Transposition.** Copying B^{T} once ($O(n^2)$ work) turns every inner product into the product of two contiguous rows.
- **Cache blocking (tiling).** The i-k-j loops are executed over square blocks of side b, so that a block of A, B and C is reused many times while it is in the cache (Algorithm 2) [5, 6]. The best b depends on the cache sizes and must be found experimentally.

In all four methods the products contributing to each C_{ij} are added in increasing order of k, so they perform the same floating-point operations in the same order and must produce bit-identical results.

Algorithm 1 Baseline i-j-k multiplication (Assignment 1)

```
1: for i ← 1 to n do
2:     for j ← 1 to n do
3:         s ← 0
4:         for k ← 1 to n do
5:             s ← s + A_ik B_kj
6:         end for
7:         C_ij ← s
8:     end for
9: end for
```

Algorithm 2 Cache-blocked (i-k-j inside blocks of side b) multiplication

$C \leftarrow 0$
for $i_0 \leftarrow 1$ **to** n **step** b **do**
  for $k_0 \leftarrow 1$ **to** n **step** b **do**
    for $j_0 \leftarrow 1$ **to** n **step** b **do**
      for $i \leftarrow i_0$ **to** $\min(i_0 + b - 1, n)$ **do**
        for $k \leftarrow k_0$ **to** $\min(k_0 + b - 1, n)$ **do**
          for $j \leftarrow j_0$ **to** $\min(j_0 + b - 1, n)$ **do**
            $C_{ij} \leftarrow C_{ij} + A_{ik} B_{kj}$
          end for
        end for
      end for
    end for
  end for
end for

2.2. Sparse matrices and the CSR format

The Compressed Sparse Row (CSR) format stores, for a matrix with nnz nonzeros, three arrays: the values and column indices of the nonzeros, row by row, and an array of $n + 1$ row pointers [3, 4]. With 64-bit values, 32-bit column indices and 64-bit row pointers it needs $12\,nnz + 8(n + 1)$ bytes instead of $8n^2$.

The product of two CSR matrices is computed with Gustavson's row-by-row algorithm [2] (Algorithm 3): row i of C is the combination of the rows of B selected by the nonzeros of row i of A, accumulated in a dense array of length n with a marker that records which columns were touched. Its work is proportional to the number of scalar multiply-adds $\sum_{(i,k):A_{ik}\neq 0} nnz(B_{k,:})$, plus the cost of producing the nonzeros of C, instead of n^3.

Algorithm 3 Gustavson's CSR $\times$ CSR product with a dense accumulator

$marker[1..n] \leftarrow -1$
for $i \leftarrow 1$ **to** n **do**
  $touched \leftarrow \emptyset$
  for each nonzero A_{ik} of row i of A **do**
    for each nonzero B_{kj} of row k of B **do**
      if $marker[j] \neq i$ **then**
        $marker[j] \leftarrow i$; append j to $touched$; $acc[j] \leftarrow A_{ik} B_{kj}$
      else
        $acc[j] \leftarrow acc[j] + A_{ik} B_{kj}$
      end if
    end for
  end for
  sort $touched$; append $(j, acc[j])$ for $j \in touched$ to row i of C
end for

Fill-in. The product of two sparse matrices is usually denser than its factors. If the nonzeros are placed uniformly at random with density d, an entry of C is zero only if none of the n products $A_{ik}B_{kj}$ is nonzero, so the expected density of C is about $1 - (1 - d^2)^n$, which approaches 1 as soon as $d^2 n$ is of order one. In contrast, the product of two banded matrices with half-bandwidth w is banded with half-bandwidth $2w$, so its density stays about twice the input density. For this reason the density alone does not determine the cost of a sparse product [4].

3. Methodology

3.1. Implementations

All implementations are our own C code, compiled with GCC 16.2.0 and `-O2 -std=c11` (Table 3.1). SciPy is used only as a reference to check the results of the sparse product. All runs are single-threaded.

Table 3.1: Methods compared. All use 64-bit floating-point values.

Method	What changes with respect to the baseline	Storage
`naive`	Baseline: the validated C implementation of Assignment 1, unchanged	Contiguous row-major arrays
`ikj`	Loop reordering: innermost loop over j	Same
`transpose`	B^T is built inside the timed region, then i-j-k over two rows	Same, plus a copy of B^T
`tiled`	Cache blocking with block side b selected experimentally	Same
`spgemm`	Gustavson's CSR × CSR product (Algorithm 3), own code	CSR (int64 row pointers, int32 columns, float64 values)
`convert`	Dense → CSR conversion of A and B (count, then fill)	Dense input, CSR output

3.2. Experiments and inputs

1. **Block tuning.** The `tiled` method with $b \in \{16, 32, 64, 128, 256\}$ for $n = 1024$ and 2048. The block with the lowest median at $n = 2048$ is used in every later tiled run.
2. **Dense methods (Part A).** The four dense methods for $n \in \{256, 512, 1000, 1024, 2048, 4096\}$. The inputs are generated exactly as in Assignment 1 (NumPy PCG64 with seed $42 + n$, uniform values in $[0, 1)$); the SHA-256 hashes of the five sizes shared with Assignment 1 are identical, so the baseline is run on the same data.
3. **Density sweep (Part B).** $n = 2048$, densities $d \in \{50, 10, 5, 1, 0.5, 0.1, 0.01\}\%$ and two structures: *uniform*, where $nnz = \text{round}(d \cdot n^2)$ positions are drawn without replacement among all n^2 positions, and *banded*, where they are drawn among the positions of the narrowest band $|i - j| \leq w$ that can hold them. Values are in $(0, 1]$, so no stored value is zero. For each input, three measurements are taken: `spgemm`, the dense `tiled` product of the same matrices, and `convert`.

4. **Size scaling (Part B).** Uniform structure, $d = 0.1\%$, $n \in \{1024, \ldots, 32768\}$, same three measurements.

The seeds of every sparse matrix and the SHA-256 hash of every input file are recorded in `inputs_manifest.csv`.

3.3. Correctness checks

`verify.py` runs 74 tests before the benchmark. Every dense method (and `tiled` with $b = 16, 64, 256$) is checked on known 1×1 and 2×2 products, $AI = A$ and $A\,0 = 0$ (exact comparison), and on random matrices with $n \in \{1, 2, 3, 17, 64, 100, 129\}$ (129 is not a multiple of any block size) against `numpy.matmul` with rtol = atol = 10^{-12}, the tolerance justified in Assignment 1 [1]. The outputs are also compared bit by bit with the baseline. The CSR product is checked on $I \times B = B$, on a matrix with empty rows, and on five random uniform and banded cases against `scipy.sparse` (same sparsity pattern, values within the tolerance) and bit by bit against the dense `ikj` product. During the benchmark, every multiplication returns a checksum of the result, which must be equal for every repetition and every method that multiplies the same input.

3.4. Measurement protocol

The protocol of Assignment 1 is reused:

- **Timer boundaries.** Dense methods: the multiplication including the allocation (and zeroing) of C and, for `transpose`, the transposition. `spgemm`: the CSR product including the allocation of the result and the sorting of the column indices of each row; reading the files is not timed. `dense` in Part B: the tiled product only (the expansion of the CSR inputs to dense arrays is not timed). `convert`: the dense→CSR conversion of A and B. Conversion is therefore measured separately and added explicitly when stated.
- **Very short configurations.** The shortest measurements are the CSR products at 0.01% density (about 30 μs), still about 300 times longer than the sub-microsecond resolution of the monotonic timer (`clock_gettime(CLOCK_MONOTONIC)`). Every multiplication is therefore timed individually instead of in batches; the higher relative IQR of these configurations is reported in Chapter 4.
- **Repetitions.** Each configuration runs in a fresh process with 1 warm-up and 5 measured repetitions; configurations whose predicted time (previous size of the same series × cubic growth) exceeds 20 s use 3 repetitions and no warm-up. Medians and interquartile ranges (IQR) are reported [9].
- **Budgets.** 300 s per multiplication and 4 GiB per process. Configurations whose

predicted array memory exceeds the budget are not run and are recorded as such; after a timeout the larger sizes of that series are not attempted.

- **Memory metric.** Peak working set of the process (Win32 `GetProcessMemoryInfo`).
- **Environment control.** Every process is pinned to logical processor 12, a performance core of the hybrid Intel Core Ultra 7 155H, with `HIGH` priority, as justified in Assignment 1 [1]. A test with two CPU-bound processes showed that logical processor 13 shares the same physical core (each process becomes about twice as slow when both run). For every configuration the harness therefore records whether the laptop was on AC power, the mean utilisation of the whole system and of processor 13, and the effective clock of processor 12 (Windows performance counter *% Processor Performance*, sampled every 0.5 s), and refuses to start on battery. Every configuration starts after the same idle period of 10 s, so that all of them begin in similar thermal conditions.

3.5. Environment

Table 3.2: Hardware and software environment of the reported run.

Item	Value
CPU	Intel Core Ultra 7 155H (6 P-cores with 2 threads each, 8 E-cores, 2 LP E-cores; 24 MB L3)
RAM	16 GB
Operating system	Windows 11 (build 26200.9711), *Best performance* power mode, on AC power
Compiler	GCC 16.2.0 (MinGW-w64 UCRT), `-O2 -std=c11`
Python	3.13.0, NumPy 2.2.6 (inputs), SciPy 1.16.3 (reference only)
CPU affinity	Logical processor 12 (performance core), `HIGH` priority, single thread

4. Results

All values come from `results/raw_measurements.csv`, `runs.csv`, `summary.csv`, `sparse_comparison.csv` and `break_even.csv`, produced by `analyze.py`.

4.1. Correctness and execution conditions

The 74 correctness tests passed. All dense methods produced results bit-identical to the baseline, and the CSR product produced exactly the same sparsity pattern and values as `scipy.sparse` and the same bits as the dense product. During the benchmark, every configuration produced a single checksum, and the checksums of `spgemm` and of the dense product of the same input were identical.

Every configuration of the reported run was executed on AC power. The median utilisation of the rest of the system was 0% (maximum 9.4%), and the utilisation of logical processor 13, which shares the physical core, exceeded 10% only once (12.4%, `transpose` with $n = 1024$). The IQR is below 3% of the median in 77 of the 89 completed configurations. The largest relative IQRs (up to 22%) occur in the shortest configurations, the CSR products at 0.01% density (about 30 μs) and the conversions for $n = 1024$ (about 2 ms), where small absolute fluctuations are large in relative terms.

Two earlier complete or partial runs were discarded and are kept in `results/`: the first one was executed on battery power, which made the long configurations up to twice as slow, and the second one under the *Balanced* power mode. These problems motivated the environment checks described in Chapter 3; the reported run used the *Best performance* power mode on AC power.

4.2. Part A: block size

Table 4.1: Tiled method: median time (s) for different block sizes (5 repetitions).

n	$b = 16$	$b = 32$	$b = 64$	$b = 128$	$b = 256$
1024	0.243	0.205	0.219	0.204	0.201
2048	2.028	1.730	**1.678**	1.740	2.042

For $n = 2048$ the blocks of 32, 64 and 128 are within 4% of each other, while 16 and 256 are about 20% slower (Table 4.1, Figure 4.1). The block $b = 64$ had the lowest median

and was used in every later tiled run; in the two discarded runs the best block was 128, so 64 and 128 should be regarded as equivalent on this machine.

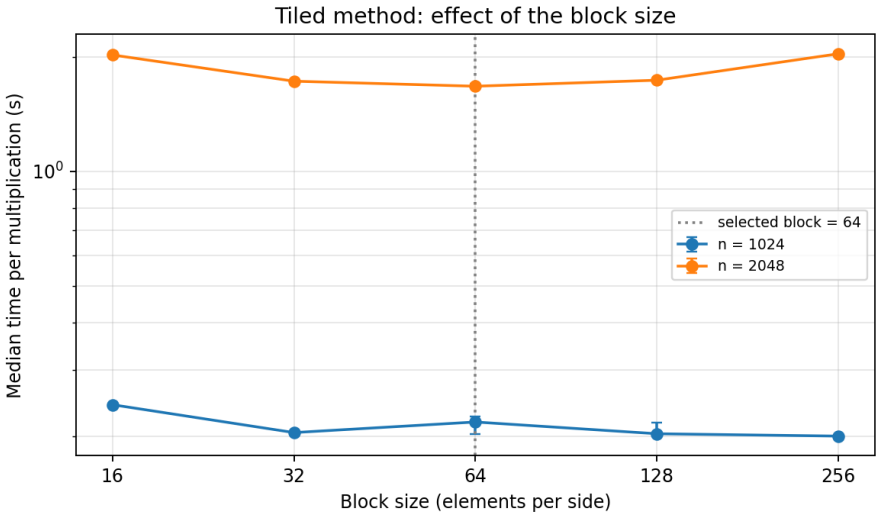


Figure 4.1: Tiled method: median time versus block size (bars: IQR).

4.3. Part A: dense methods

Table 4.2: Dense methods: median time in seconds (IQR below 3% of the median except where indicated) and speedup over the baseline. All runs: 5 repetitions, except $n = 4096$ for `ikj` and `transpose` (3 repetitions).

	naive	**ikj**		**transpose**		**tiled** ($b = 64$)	
n	time	time	speedup	time	speedup	time	speedup
256	0.0117	0.00293	4.0	0.00587	2.0	0.00283	4.1
512	0.111	0.0228 (IQR 4%)	4.9	0.0541	2.0	0.0236	4.7
1000	0.502	0.183	2.7	0.427	1.2	0.166	3.0
1024	2.140	0.196	10.9	0.502 (IQR 7%)	4.3	0.196	10.9
2048	86.56	4.122	21.0	5.487	15.8	1.659	52.2
4096	timeout ($>$ 300 s)	35.52	–	44.63	–	15.31	–

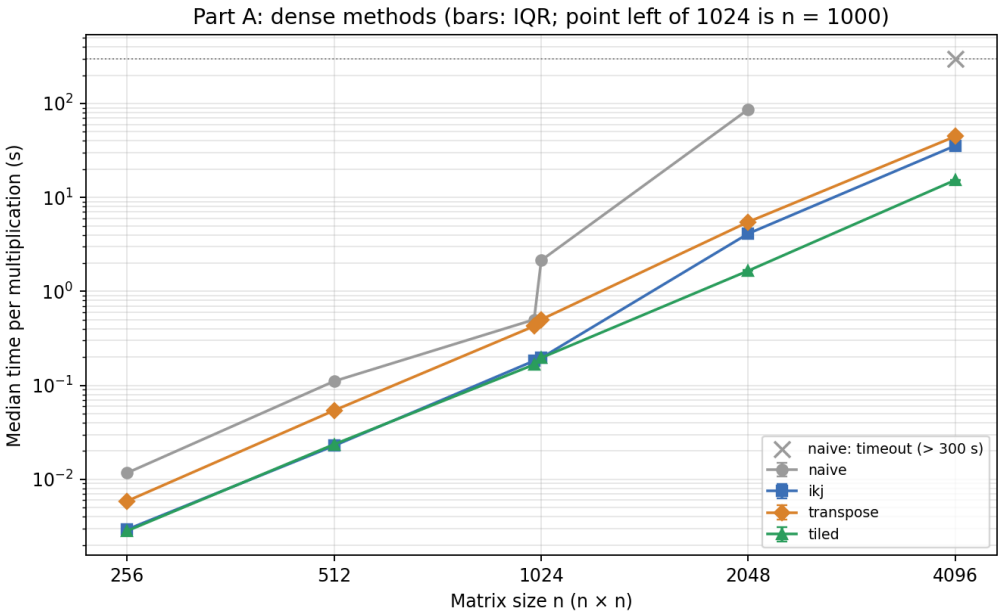


Figure 4.2: Part A: median time versus n (log–log; bars: IQR; the point just to the left of 1024 is $n = 1000$).

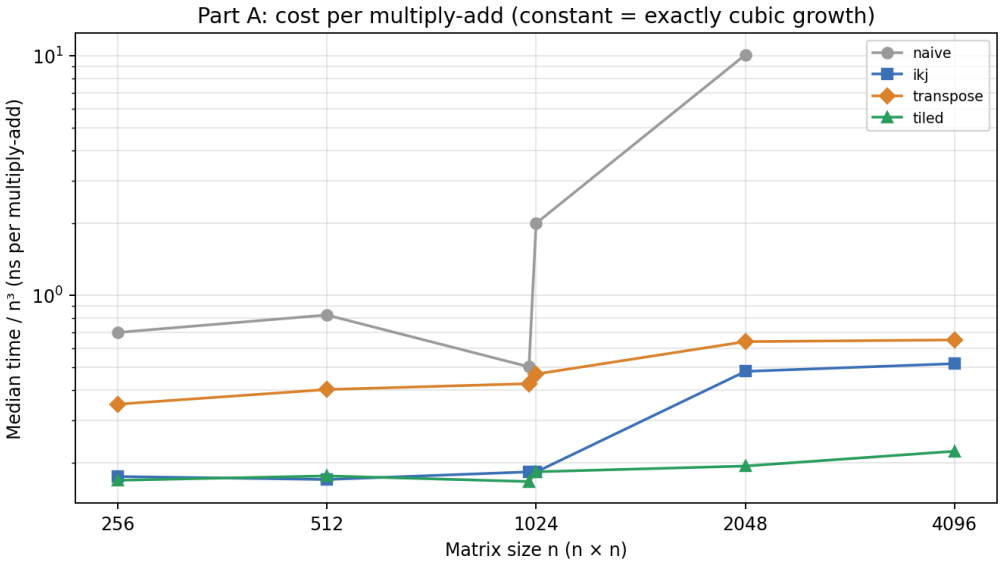


Figure 4.3: Part A: median time divided by n^3. A horizontal line means exactly cubic growth.

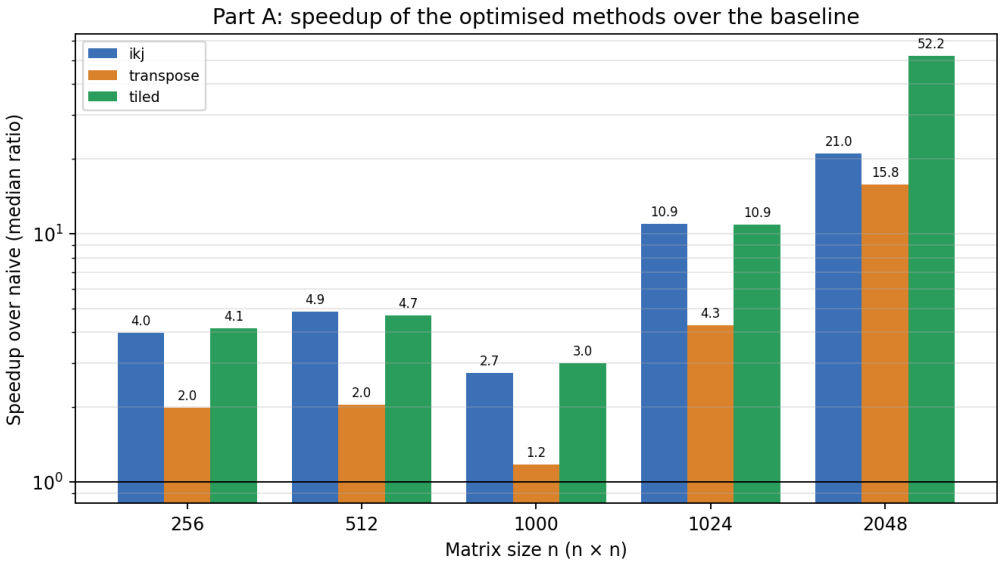


Figure 4.4: Part A: speedup of each optimised method over the baseline (median ratio).

Table 4.2 and Figures 4.2 to 4.4 show the main results. The cost per multiply-add of the tiled method stays between 0.17 and 0.22 ns for all sizes, and its time grows by 9.2 from $n = 2048$ to 4096 (cubic model: 8). The `ikj` method has the same cost as `tiled` up to $n = 1024$ (0.17–0.18 ns) but rises to 0.48–0.52 ns for $n \geq 2048$. `transpose` costs 0.35–0.65 ns. The baseline goes from 0.70 ns at $n = 256$ to 10.1 ns at $n = 2048$, and its time multiplies by 40 from $n = 1024$ to 2048. Between $n = 1000$ and 1024 (cubic model: 1.07) the baseline becomes 4.3 times slower, `ikj` 1.07 times and `tiled` and `transpose` 1.18 times. Peak memory is about $24n^2$ bytes for every method except `transpose`, which needs the extra copy of B^{T} (516 MiB against 388 MiB for $n = 4096$).

Limits. The largest attempted size was $n = 4096$ for all four methods. The baseline did not finish its first multiplication with $n = 4096$ within the 300 s limit, so its largest feasible tested size is $n = 2048$; for `ikj`, `transpose` and `tiled` the largest feasible tested size is $n = 4096$. No dense configuration came close to the 4 GiB memory budget (maximum 516 MiB).

Figure 4.5 shows the effective clock of the pinned core, recorded every 0.5 s for configurations that lasted long enough to be sampled. The optimised methods ran at 4.0–4.8 GHz, but the baseline ran at 3.1 GHz for $n = 2048$ and 1.7 GHz during the timed-out run with $n = 4096$, although all of them used the same core, power mode and temperature conditions.

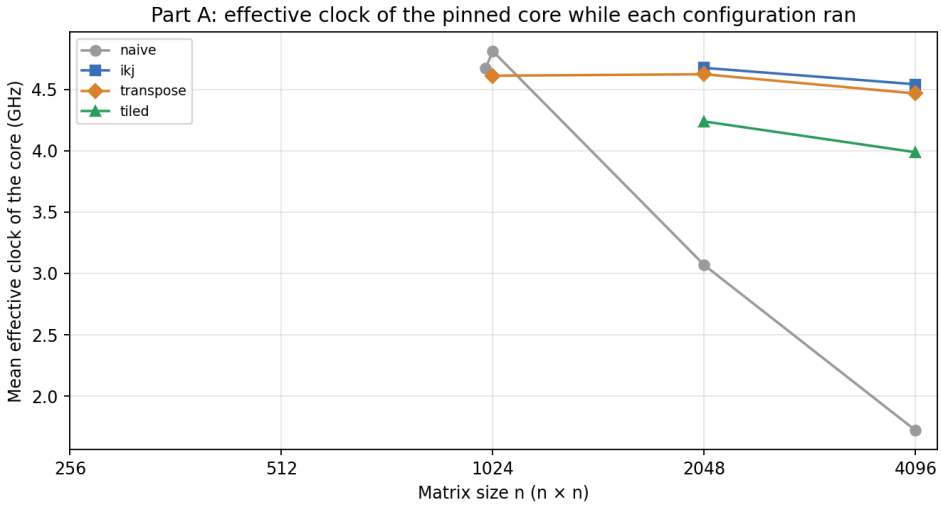


Figure 4.5: Part A: mean effective clock of the pinned core during each configuration (Windows counter *% Processor Performance*; only configurations long enough to be sampled are shown).

4.4. Part B: density and structure

Table 4.3: Part B, $n = 2048$: nonzeros of each input and of the product, median times in seconds, and speedup of the CSR product over the dense tiled product of the same matrices (without and with the conversion of A and B).

		Nonzeros			**Median time (s)**			**Speedup over dense**	
Structure	d (%)	$nnz(A)$	$nnz(C)$	density C (%)	spgemm	dense	convert	spgemm	incl. conv.
uniform	0.01	419	75	0.002	0.0000308	1.622	0.0111	52 660	146
uniform	0.1	4 194	8 661	0.21	0.000199	1.604	0.0117	8 050	135
uniform	0.5	20 972	209 800	5.0	0.00882	1.623	0.0116	184	80
uniform	1	41 943	777 200	18.5	0.0393	1.611	0.0102	41	33
uniform	5	209 715	4 169 000	99.4	0.279	1.637	0.0133	5.9	5.6
uniform	10	419 430	4 194 304	100	0.307	1.634	0.0175	5.3	5.0
uniform	50	2 097 152	4 194 304	100	1.882	1.637	0.0448	0.87	0.85
banded	0.01	419	87	0.002	0.0000351	1.552	0.0109	44 220	143
banded	0.1	4 194	6 562	0.16	0.0000993	1.583	0.0111	15 940	142
banded	0.5	20 972	42 290	1.0	0.000629	1.639	0.0121	2 605	129
banded	1	41 943	83 380	2.0	0.00138	1.620	0.0120	1 173	121
banded	5	209 715	417 000	9.9	0.0185	1.618	0.0123	87	52
banded	10	419 430	817 900	19.5	0.0585	1.610	0.0129	27.5	22.5
banded	50	2 097 152	3 476 000	82.9	1.282	1.623	0.0181	1.27	1.25

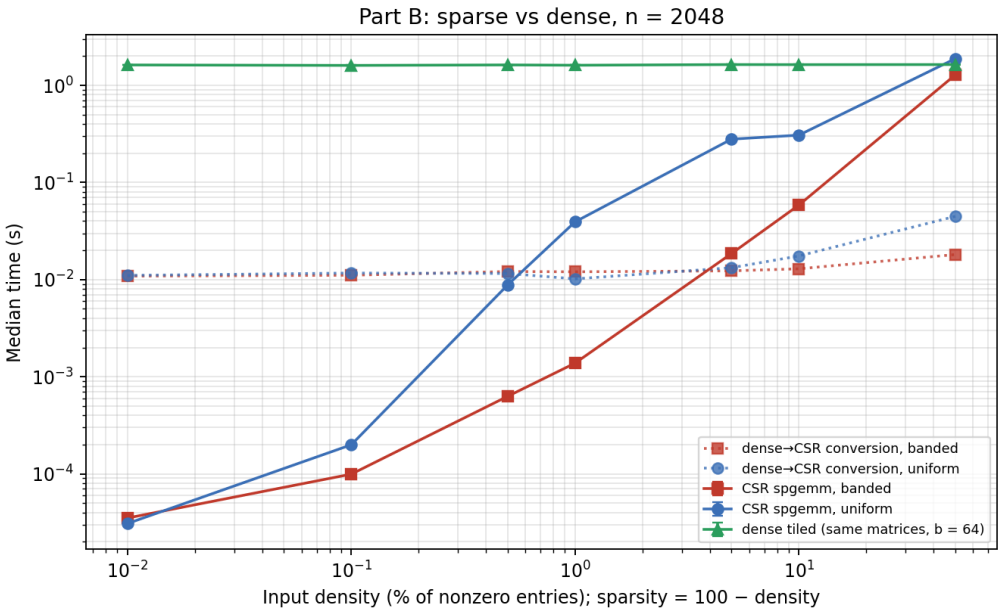


Figure 4.6: Part B, $n = 2048$: median time of the CSR product (solid), of the dense $\to$ CSR conversion (dotted) and of the dense tiled product of the same matrices, versus input density.

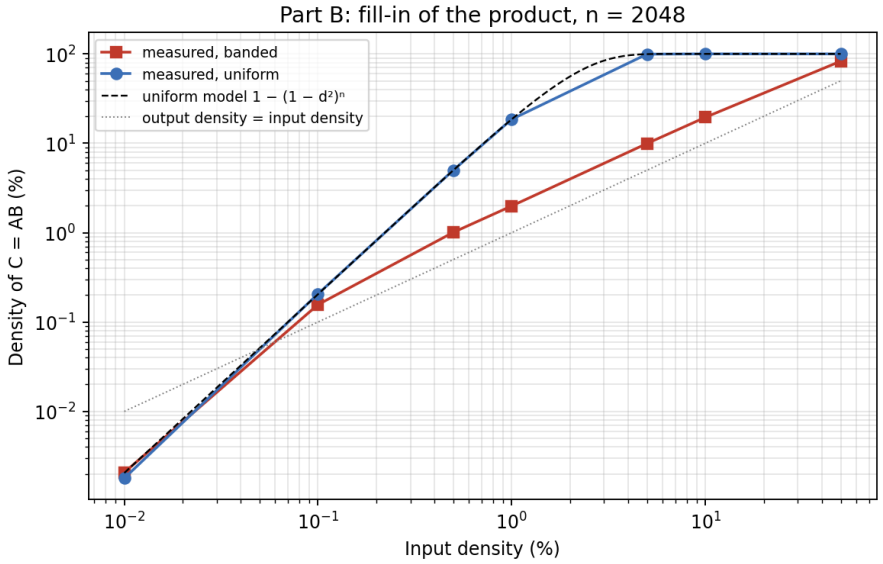


Figure 4.7: Part B, $n = 2048$: density of the product versus density of the inputs, with the uniform model $1 - (1 - d^2)^n$.

The dense product takes 1.55–1.64 s for every input, independently of the number of zeros (Table 4.3, Figure 4.6). The CSR product is faster for every banded input and for uniform inputs up to 10%, and slower for the uniform input with 50%; interpolating in log–log scale, the break-even density for the uniform structure is 44% (43% when the conversion is included), and no break-even exists in the tested range for the banded structure. The conversion of A and B costs 10–45 ms, nearly independent of the density, which is more than the CSR product itself for every density up to 0.5% (uniform) or 1% (banded).

Figure 4.7 shows that the measured density of the product of uniform matrices matches the model $1 - (1 - d^2)^n$: it is 18.5% for $d = 1\%$ and 99.4% for $d = 5\%$. The product of banded matrices stays at about twice the input density. With the same input density of 5%, the uniform product has ten times more nonzeros and takes 15 times longer than the banded one. The cost per scalar multiply-add of the CSR product was 0.6–1.6 ns for banded inputs with $d \geq 1\%$, but 13–46 ns for uniform inputs with $0.5\% \leq d \leq 5\%$; at the lowest densities it is dominated by the per-row work and is not meaningful.

Peak memory was 4.5–18 MiB for the CSR product with $d \leq 1\%$ against 100 MiB for the dense product, but for uniform inputs with $d \geq 5\%$ the CSR product needed 73–100 MiB, as much as the dense one, because the result is practically dense and CSR needs 12 bytes per stored value instead of 8.

4.5. Part B: size scaling

Table 4.4: Part B, uniform structure with $d = 0.1\%$: median times (s), product density and peak memory (MiB).

n	$nnz(A)$	density C (%)	spgemm	dense	convert	mem. spgemm	mem. dense
1024	1 049	0.11	0.0000353	0.192	0.0022	4.4	28
2048	4 194	0.21	0.000208	1.594	0.0114	4.6	100
4096	16 777	0.41	0.00199	13.32	0.0480	5.7	388
8192	67 109	0.82	0.0211	110.2	0.195	12.2	1 539
16384	268 435	1.63	0.209	not run*	not run*	60.8	–
32768	1 073 742	3.22	2.090	not run*	not run*	426	–

* Predicted array memory above the 4 GiB budget (dense: 6 GiB for $n = 16384$, 24 GiB for $n = 32768$).

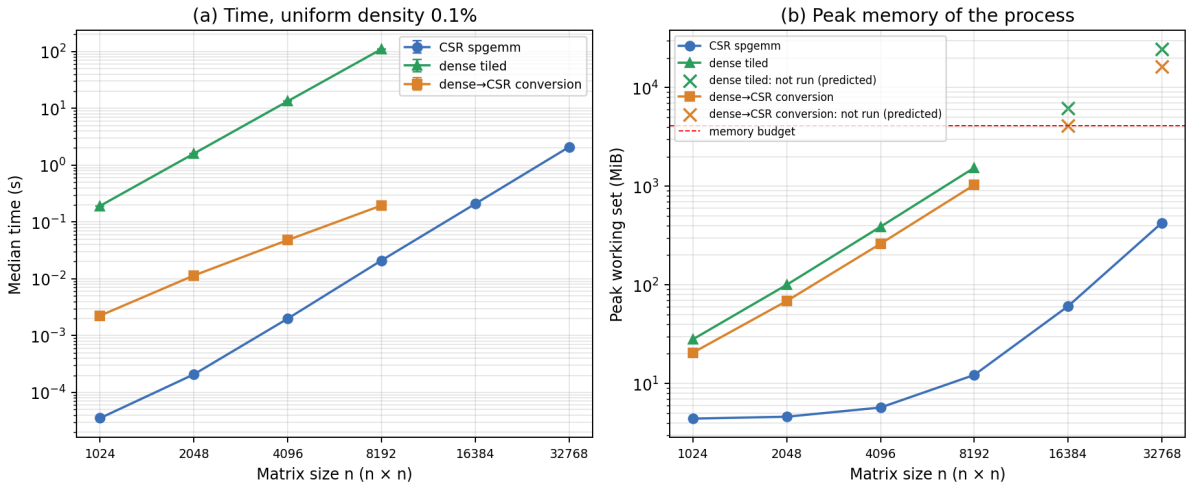


Figure 4.8: Part B, uniform $d = 0.1\%$: (a) median time and (b) peak memory versus n. Crosses: predicted memory of the configurations that were not run because they exceed the budget.

For a fixed density of 0.1%, the CSR product was 5 200 to 7 700 times faster than the dense product for $n \leq 8192$ (Table 4.4, Figure 4.8). Its time grows by about 10 per doubling of n from $n = 2048$ onwards (6 between 1024 and 2048), because both the number of multiply-adds ($\propto d^2 n^3$) and the density of the product grow with n. The largest feasible size was $n = 8192$ for the dense product and conversion, limited by memory, and $n = 32768$ (the largest tested) for the CSR product.

5. Discussion

5.1. Effect of the dense optimisations (H1–H3)

H1 is supported. The three optimisations were faster than the baseline at every size, and the advantage grew with n: from 2–4 times at $n = 256$ to 21 (`ikj`), 16 (`transpose`) and 52 (`tiled`) times at $n = 2048$. The only exception to the growth is $n = 1000$, where the baseline is not penalised by the power-of-two effect and the gains are smaller (1.2–3.0 times). Because every method performs the same operations in the same order and produces the same bits, these differences come only from the memory access pattern.

The cost per multiply-add explains the differences (Figure 4.3). The innermost loop of `ikj` streams a row of B and a row of C; for $n \leq 1024$ all of B (at most 8 MiB) fits in the 24 MB L3 cache, but for $n = 2048$ it occupies 32 MiB, so every pass of the k loop reads B from main memory and the cost per operation nearly triples. The tiled method works on 64×64 blocks (32 KiB each), so the block of B being used stays in the private caches and is reused for 64 rows of C; its cost per operation is almost independent of n, which is exactly the behaviour expected from blocking [5, 6]. `transpose` reads two contiguous rows, but its inner loop is a reduction into a single accumulator, a chain of dependent additions that cannot be overlapped, whereas the `ikj` and `tiled` inner loops update independent elements of C; this is a plausible reason for its higher cost even when the data fits in cache, but it was not verified with hardware counters.

H2 is supported. `tiled` was the fastest method for $n \geq 2048$ and the only one whose cost per operation stayed constant up to $n = 4096$, where it was 2.3 times faster than `ikj` and 2.9 times faster than `transpose`. For $n \leq 1024$, where B fits in the L3 cache, `ikj` and `tiled` were equivalent (within 10%, and within 1% at $n = 1024$), so blocking only pays off once the working set exceeds the cache. The block size had a broad optimum (32–128) on this processor; very small blocks add loop overhead and very large blocks no longer fit in the private caches.

H3 is supported. The baseline was 4.3 times slower at $n = 1024$ than at $n = 1000$, as in Assignment 1 [1], because the column stride of a power-of-two matrix maps the elements of a column to few cache sets. All the optimised methods avoid traversing columns, and their ratio between these two sizes is 1.07–1.18, close to the cubic prediction of 1.07.

Clock frequency and the baseline. The baseline at $n = 2048$ ran at 3.1 GHz and the optimised methods at 4.0–4.8 GHz on the same core (Figure 4.5). The power management of the processor lowered the clock when the core was mostly stalled waiting for memory, which is the situation of the column-wise traversal of the baseline. Part of the 52-times

speedup of `tiled` at $n = 2048$ is therefore a frequency effect; scaling the baseline time by the ratio of the clocks (3.07/4.24) still leaves a speedup of about 38. The same baseline code and inputs took 32.2 s in Assignment 1 (14 October 2026) and 86.6 s in this study; running the unmodified Assignment 1 binary on the same day as this study also took 93 s, so the difference is not due to the code. Windows and Intel graphics driver updates were installed between the two dates, but the exact cause could not be identified. For this reason absolute times should only be compared within the same study; all comparisons in this report were measured in a single run under the same conditions.

5.2. When does the sparse representation pay off? (H4–H5)

H4 is supported, and it shows that density alone does not determine the performance. The dense product does not benefit from zeros, so its time is constant, while the CSR product's work depends on the number of useful multiply-adds and on the size of the result. For uniform matrices both grow very fast: the result is 99% dense from $d = 5\%$, so above that point the CSR product produces a dense matrix through a slower, indirect path, and its advantage vanishes at about 44%. Banded matrices with the same density produce only twice as many nonzeros, their rows of B are close to each other, and the touched columns of the accumulator form a narrow range, which explains their much lower cost per operation (0.6–1.6 ns against 13–46 ns) and why the CSR product was still faster at 50%. With the same 5% density, the uniform product was 15 times slower than the banded one. The break-even density depends on n as well, because the fill-in grows with d^2n.

The conversion from dense to CSR costs $O(n^2)$ regardless of the density (10–45 ms for $n = 2048$). It does not change the break-even density, but for very sparse inputs it dominates: the speedup over the dense product falls from more than 40 000 to about 140 when the conversion is included. In practice, sparse data should be generated or read directly in a sparse format rather than converted from a dense array.

H5 is supported. At $d = 0.1\%$ the dense product needs $24n^2$ bytes and was limited to $n = 8192$ by the 4 GiB budget (1.5 GiB and 110 s), while the CSR product handled $n = 32768$ in 2.1 s and 426 MiB. The advantage is not unlimited: the density of the product grew from 0.1% to 3.2% over this range, so for even larger n at the same input density the result would eventually become dense.

5.3. Observed bottlenecks

The measurements point to the following bottlenecks, in order of impact:

- **Memory access of the baseline.** The column-wise traversal of B is the main

bottleneck of the dense baseline: its cost per operation grows from 0.7 to 10 ns as B leaves the caches, it suffers conflict misses for power-of-two sizes, and the processor lowers its clock while it waits for memory.

- **L3 capacity for loop reordering.** `ikj` becomes three times more expensive per operation when B (32 MiB at $n = 2048$) no longer fits in the 24 MB L3 cache; blocking removes this bottleneck.
- **Dependent additions in `transpose`.** Its single-accumulator reduction is plausibly the reason why it is slower than `ikj` even when all data fits in cache.
- **Fill-in and irregular access in the CSR product.** For uniform inputs the result becomes dense and the scattered updates of the accumulator cost 13–46 ns per multiply-add, so the advantage of sparse storage disappears at about 44% density.
- **Conversion.** The $O(n^2)$ dense $\rightarrow$ CSR conversion is the bottleneck for very sparse inputs.
- **Memory capacity.** Dense storage limits the problem size to $n = 8192$ within the 4 GiB budget.

5.4. Optimisation versus sparse storage

The two parts improve performance by different mechanisms. The dense optimisations do not change the amount of arithmetic: they reduce the cost of each operation, at most by a factor of about 50 here, and their benefit grows with n. Sparse storage reduces the amount of work and memory, and its benefit can be several orders of magnitude, but only while the inputs and the result stay sparse; it depends on the density, on the structure of the nonzeros, on n and on the conversion costs, and it can be slower than a good dense method for dense results.

5.5. Limitations

- All measurements come from one laptop and one core; no hardware performance counters (cache, TLB misses) were collected, so the explanations based on the memory hierarchy are inferred from the timing patterns and from the clock measurements.
- The power management of the processor changes the clock during memory-bound phases. The clock was recorded, but not fixed, so the comparison with the baseline mixes memory effects and frequency effects.
- The absolute times of the baseline changed between Assignment 1 and this study for reasons that could not be identified (see above).
- Only one sparse format (CSR), one algorithm (Gustavson with a dense accumulator) and two synthetic structures were tested; real sparse matrices have other structures.

No optimised sparse library was benchmarked.

- Only one size ($n = 2048$) was used for the density sweep, and only uniform matrices with $d = 0.1\%$ for the scaling study.
- The blocked method uses a single level of blocking with square blocks and no vectorisation; optimised BLAS libraries use several levels and are much faster [6].

6. Conclusions

Within the tested implementations, inputs and machine, this study establishes that:

- Changing only the order of the same operations makes the dense multiplication up to 52 times faster ($n = 2048$). Cache blocking with blocks of 32–128 elements is the most effective technique for large matrices: its cost per operation stays almost constant up to $n = 4096$, so it follows the cubic model, while loop reordering degrades once a matrix no longer fits in the L3 cache.
- The optimisations remove the power-of-two penalty of the baseline.
- Memory-bound code can also lower the clock frequency chosen by the processor, which amplifies the cost of a poor access pattern.
- A CSR product is faster than a dense product of the same matrices below a break-even density of about 44% for uniformly random nonzeros at $n = 2048$, and in the whole tested range for banded nonzeros. The difference is caused by fill-in: uniform sparse matrices produce dense results quickly. Density alone does not determine the result.
- The conversion from dense to CSR costs $O(n^2)$ and dominates for very sparse inputs.
- For very sparse matrices, sparse storage makes much larger problems feasible: at 0.1% density, $n = 32768$ took 2.1 s and 426 MiB, while the dense product was limited to $n = 8192$ by memory.

Open questions for future work are the direct measurement of cache misses and clock behaviour, other sparse formats and real matrices, and the combination of these techniques with the parallel execution of Assignment 3.

AI Usage Statement

In accordance with the course policy, a generative AI assistant was used to support the planning of the experiments, the development and debugging of the benchmark code and scripts, and the language revision of this report. All the measurements were executed on the author's computer, and the author checked the code, the results and the references and is responsible for the conclusions.

Reproducibility Materials

All materials are in the folder `assignment2` of the repository https://github.com/Claudia1771/bigdata-matrix-multiplication:

- `src/dense.c` and `src/sparse.c`: the implementations (the baseline is the code of `assignment1`).
- `config.json`: seeds, sizes, densities, structures, repetitions, budgets, flags and CPU affinity.
- `scripts/`: `generate_inputs.py`, `verify.py`, `run_experiment.py`, `analyze.py` and `common.py`.
- `results/`: `inputs_manifest.csv` (seeds and SHA-256 hashes), `correctness.csv`, raw measurements, run conditions, summaries, figures, and the two discarded runs.
- `README.md`: exact build and run commands and dependency versions.

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